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Methods in Ecology and Evolution

CORRECTION

Correction to 'An evaluation of multi-species occupancy models with correlated species occurrences'

Dorazio, R. M., Li, P., Xiao, L., & Kachel, S. M. (2025) An evaluation of multi-species occupancy models with correlated species occurrences. *Methods in Ecology and Evolution*, 16(11), 2681–2693. https://doi.org/10.1111/2041-210X.70168

The full conditional distribution of $Z_{i,j}$ that was used to fit the multivariate probit (MP) occupancy model (Appendix S1) failed to properly account for the correlations in occurrence between species.

The correct full conditional distribution of $Z_{i,j}$ is

$$Z_{i,j} \mid \cdot \sim \begin{cases} \text{Bernoulli}(1) & \text{if } \sum_{k=1}^{K_i} Y_{i,k,j} \neq 0 \\ \text{Bernoulli}\left(\theta_{i,j}\right) & \text{if } \sum_{k=1}^{K_i} Y_{i,k,j} = 0 \end{cases}$$

where

$$\theta_{i,j} = \frac{\Phi\left(\frac{\mu_w}{\sigma_w}\right) \prod_{k=1}^{K_i} \left(1 - p_{i,k,j}\right)}{1 - \Phi\left(\frac{\mu_w}{\sigma_w}\right) + \Phi\left(\frac{\mu_w}{\sigma_w}\right) \prod_{k=1}^{K_i} \left(1 - p_{i,k,j}\right)}$$

and

$$\mu_w = \mu_{i,j} + \sigma_j' \, \boldsymbol{\Sigma}_{-j,-j} \, \left(w_{i,-j} - \mu_{i,-j}\right)$$

$$\sigma_w^2 = \sigma_{j,j} + \sigma_j' \, \boldsymbol{\Sigma}_{-j,-j} \, \sigma_j$$

$w_{i,-j}$ = all elements of $w_i = \left(w_{i,1}, \ldots, w_{i,m}\right)'$ except $w_{i,j}$

$\mu_i = B' x_i$

$\mu_{i,-j}$ = all elements of μ_i except $\mu_{i,j}$

$\boldsymbol{\Sigma}_{-j,-j}$ = principal submatrix of $\boldsymbol{\Sigma}$ without its jth row and jth column

σ_j = all elements in jth column of $\boldsymbol{\Sigma}$ except $\sigma_{j,j}$

This error influenced the results of the simulation studies (Figures 1–4) and the estimated correlations in occurrence between 13 mammal species in the Sanjiangyuan Region on the Tibetan Plateau (Figure 5; Table S1). Correcting this error in our MCMC algorithm eliminated the previously reported systematic bias in estimates of correlation parameters obtained by fitting the MP model (see update of Figure 4 below). Estimated coverages of the 95% credible intervals shown in the update of Figure 4 are as follows: 0.933 (SE = 0.004) for the LF model, 0.947 (SE = 0.004) for the MP model and 0.945 (SE = 0.004) for the MvP model.

The estimated correlations in occurrence between 13 mammal species on the Tibetan Plateau also were affected by correcting the MCMC algorithm used to fit the MP occupancy model (see update of Figure 5 below). The number of estimates that differed significantly from zero and the magnitudes of these estimates increased after correcting the error in the previous MCMC algorithm. Most of these significant correlations are likely attributed to similarities in habitat preferences as opposed to direct interactions between species.

Our code, which has been updated to implement the correct full conditionals for fitting the MP occupancy model, is freely available at https://doi.org/10.5281/zenodo.17178936 and https://github.com/RobertDorazio/MultispeciesOccupancy.

We apologize for this error, and we thank Ben Augustine for bringing it to our attention.

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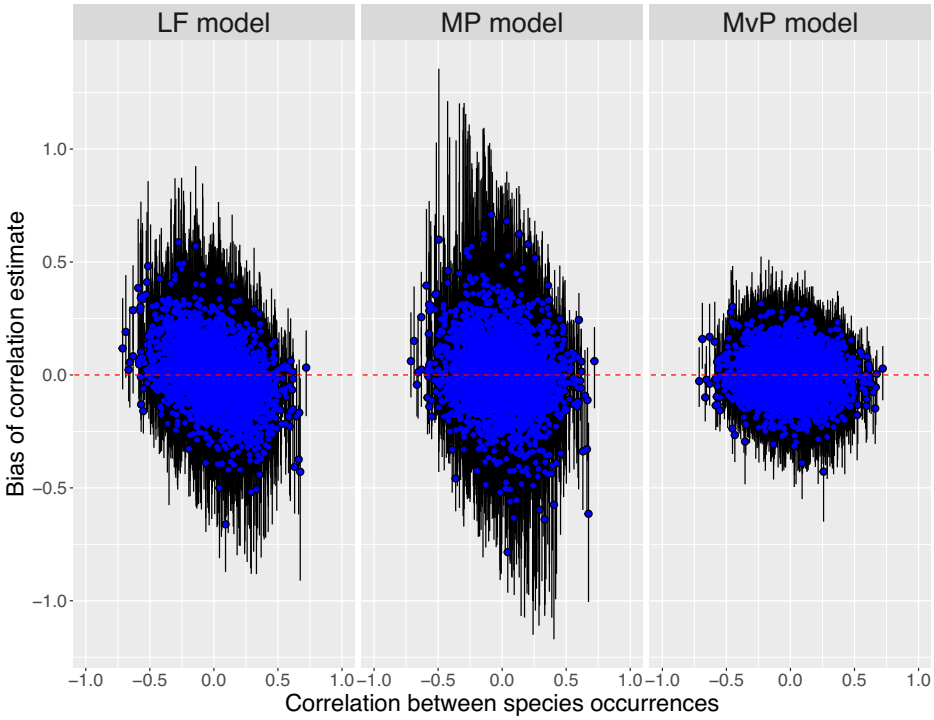


FIGURE 4 Bias (median and 95% credible interval) of correlation parameters estimated by fitting occupancy models (MP or LF) to simulated detection data or by fitting a multivariate probit regression model (MvP) to simulated species occurrences. The data and occurrences were obtained in simulations of $m = 10$ species and $n = 400$ survey locations.

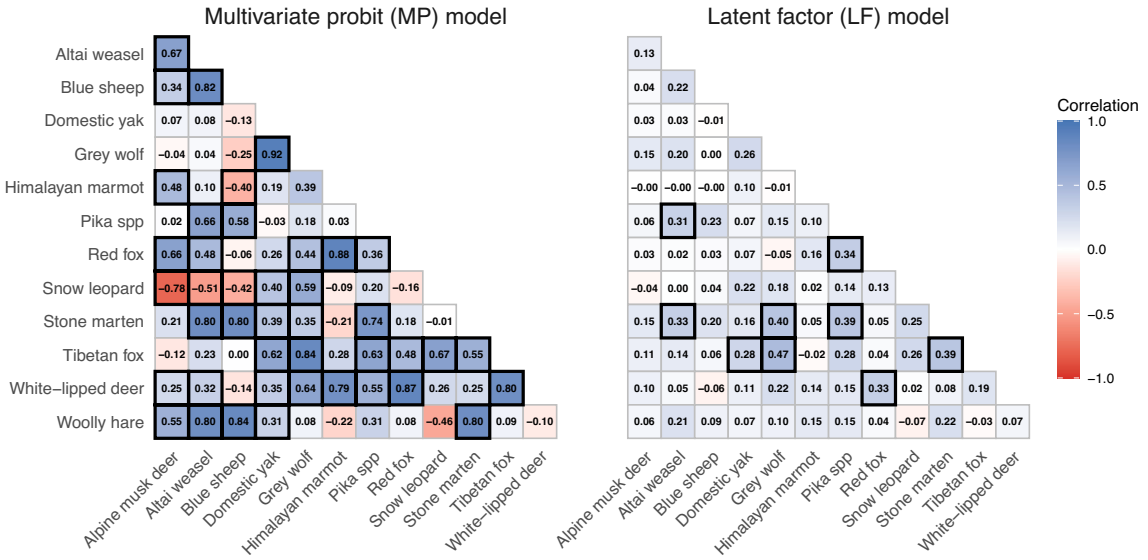


FIGURE 5 Estimated correlations in occurrence between 13 mammal species in the Sanjiangyuan Region on the Tibetan Plateau. The black border indicates correlations whose 90% credible interval did not include zero.

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